gpeyt1-35 · Secondary compression (organic)

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

A 14 ft layer of organic silt (OH, organic content 14%) under a preloaded embankment reaches the end of primary consolidation at t = 1.8 years. The strain-based coefficient of secondary compression is Cα = 0.012. The design life of the roadway is 40 years from construction. Primary settlement was 22 in.

Most nearly, the secondary settlement from end of primary to end of design life is:

A. 3.2 in.
B. 6.3 in.
C. 1.0 in.
D. 2.7 in.

Given equations / data

> Ss = Cα H0 Δlog t (Cα in strain form)

Worked solution

1. Δlog t = log(40/1.8) = 1.3468
2. Ss = 0.012 × 14 × 1.3468 = 0.2263 ft = 2.7 in.

Final answer

2.7 in.

Why the other values are wrong

- **3.2 in.** — used log t2 (started the clock at 1 yr)
- **6.3 in.** — used natural log
- **1.0 in.** — divided by (1 + e0) again although Cα is in strain form

References (Jayce's library)

- Vandevelde, ASCE PE Review — Foundation Engineering (Ss = Cα H0 Δlog t)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=IGS94AWmPWA (no creator values copied).

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